Reasoning Failure Boundaries in ECG Foundation Models Under Progressive Physiological Information Loss

First Review

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Presentation Outline

1. Introduction
2. Literature Review
3. System Design
4. Implementation and Results
5. Project Management
6. Conclusion

Background and Motivation

- ECG models can reason well when the full physiological picture is available
- But as leads disappear, the evidence available to support that reasoning also disappears
- The model may still produce a plausible answer even when its physiological evidence is becoming insufficient
- Can structured clinical knowledge help preserve grounded reasoning without replacing the missing evidence?

Core Motivation

To understand when ECG reasoning stops being trustworthy as physiological evidence disappears. And to see whether clinical knowledge can help preserve that reasoning for longer.

Problem Statement

Problem

Current ECG evaluation does not characterize the degradation of evidence-grounded clinical reasoning as physiological information decreases.

- Clinical reasoning is not evaluated across progressive levels of ECG information.
- Lead loss removes physiological evidence needed to support clinical reasoning.
- Accuracy-based and partial-lead evaluations measure whether the model remains correct, but not whether its reasoning remains grounded in the evidence still available.
- The evaluation must preserve controlled information loss, prevent reconstruction of missing evidence, and distinguish clinically supported reasoning from unsupported inference.

Project Objectives

1. To establish a controlled framework for evaluating ECG clinical reasoning under progressive physiological information loss.
2. To compare ECG-only reasoning with **Clinical Knowledge Prior (CKP)-assisted reasoning** under identical information-loss conditions.
3. To evaluate whether clinical reasoning remains grounded in the physiological evidence available at each level of information loss.
4. To characterize the **Reasoning Degradation Curve (RDC)** by measuring grounding quality and diagnostic performance under progressive ECG lead masking.
5. To determine the **Reasoning Failure Boundary (RFB)** and assess whether CKP can delay this boundary without increasing unsupported clinical inference.

Scope and Expected Outcomes

Included

- **Target users and use cases:** ECG-AI researchers and clinical AI system designers evaluating reasoning under incomplete ECG information.
- **Core modules:** ECG foundation-model adaptation, hypothesis preparation, CKP-assisted reasoning, and evidence grounding.
- **Datasets:** PTB-XL, MIMIC-IV-ECG and Chapman Shaoxing
- **Evaluation:** Grounding Quality, Reasoning Degradation Curve, AURDC, and RFB.

Excluded / Constraints

- **Features outside current phase:** Lead reconstruction, new foundation-model development, and non-ECG reasoning tasks.
- **Dataset and hardware constraints:** Disease selection depends on dataset coverage and training on available compute resources.
- **Dependencies:** Requires pretrained models, clinical evidence, and CKP construction.
- **Deployment boundaries:** Limited to research benchmarking.

Literature Review

Ref.	Method / Technology	Key Contribution	Limitation / Research Gap
[1]	**Strodthoff et al. (2021)** — PTB-XL Benchmark	Standardizes ECG model evaluation.	Does not evaluate reasoning under progressive lead loss.
[2]	**K-MERL (2025)** — Arbitrary-Lead ECG	Enables ECG representation learning with arbitrary leads.	Does not evaluate grounding during progressive lead loss.
[3]	**ECG-FM (2025)** — ECG Foundation Model	Provides strong pretrained ECG representations and benchmarking.	Does not evaluate grounded reasoning under progressive lead loss.
[4]	**GEM (2025)** — Grounded ECG Understanding	Links ECG findings to measurable evidence for grounded reasoning.	Does not study grounding degradation under progressive lead loss.

Research Gap and Proposed Contribution

Identified Gap

- Reasoning is rarely evaluated under progressive physiological information loss.
- The point at which ECG diagnosis becomes unreliable due to insufficient evidence remains unclear.
- The role of structured knowledge in delaying this failure is unexplored.

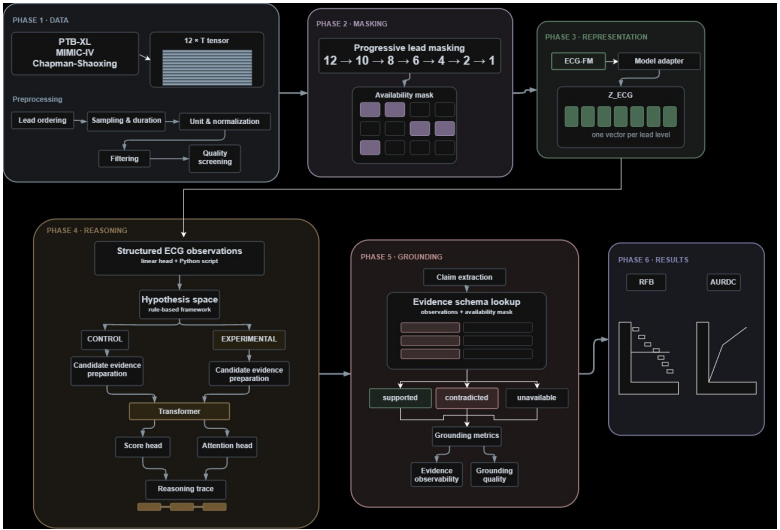
Proposed Contribution

- A controlled framework for progressive ECG information loss.
- ECG-only vs. CKP-assisted reasoning comparison.
- Area under Reasoning Degradation Curve (AURDC) and Reasoning Failure Boundary.

Existing System vs. Proposed System

Criterion	Existing System	Proposed System
Approach	Evaluates ECG models under fixed or predefined input conditions.	Evaluates reasoning across progressively reduced ECG information.
Accuracy / Quality	Focuses mainly on getting the diagnosis correct.	Evaluates diagnosis correctness and reasoning grounding.
Cost / Time	Requires conventional model evaluation at selected input conditions.	Requires repeated evaluation across information-loss levels.
Security / Privacy	Operates within existing clinical-data access constraints.	Maintains the same data-access boundaries without additional data collection.
Scalability	Generally evaluates fixed lead configurations.	Supports systematic evaluation from 12 to 1 available leads.

Proposed System Architecture



System Modules

1. **Module 1 – Data & Preparation:** Loads and standardises ECG data and prepares the 12-lead input with lead-availability masks.
2. **Module 2 – Controlled Degradation:** Progressively masks ECG leads from 12 to 1 using lead-availability masks without reconstruction.
3. **Module 3 – Representation & Observation:** Uses the selected ECG foundation model and Model Adapter to obtain standardised ECG representations and structured ECG observations.
4. **Module 4 – Reasoning:** Builds the hypothesis space and candidate evidence, then performs ECG-only and CKP-assisted reasoning using the trainable Transformer.
5. **Module 5 – Grounding:** Maps the generated reasoning claims to the Evidence Schema and verifies them against the available ECG evidence.
6. **Module 6 – Evaluation & RFB:** Measures Grounding Quality and diagnostic performance, constructs the Reasoning Degradation Curve, and derives AURDC and the Reasoning Failure Boundary (RFB).

Methodology / Workflow

1. Data Preparation
2. ECG Preprocessing & Lead Masking
3. Foundation Model & Representation
4. Hypothesis & CKP-Assisted Reasoning
5. Claim Grounding & Verification
6. Degradation Analysis & RFB

Evaluation Metrics

- Grounding Quality
- Reasoning degradation Curve
- Reasoning Failure Boundary
- Area Under the Reasoning Degradation Curve (AURDC)

Core Algorithm / Processing Logic

Algorithm: Clinical Knowledge Prior Construction

1. Obtain ECG-relevant concepts from SNOMED-CT.
2. Identify ECG findings and their associated clinical relationships.
3. Organise concepts, findings, and relationships into an ECG-specific knowledge structure.
4. Define evidence-to-disease relationships and verification rules.
5. Construct the ECG-specific Clinical Knowledge Prior (CKP).
6. Retrieve candidate-relevant CKP concepts during reasoning.

Technology Stack and Resources

Category	Selected Technology / Resource
Language & Libraries	Python, NumPy, SciPy, Pandas
Framework	PyTorch / Hugging Face Transformers
Pre-built ECG Tools	WFDB, NeuroKit2, ECG preprocessing and signal-processing utilities
Database	Not required
Dataset	PTB-XL, MIMIC-IV-ECG, Chapman-Shaoxing, SNOMED-CT
Development Tools	Jupyter / VS Code, Git, GPU-enabled environment
Hardware	GPU-enabled system, sufficient RAM and storage

Implementation Status

Component	Status	Evidence / Remarks
Requirements and Design	Completed	Objectives, scope, and architecture defined
Datasets & Input Preparation	Planned	Datasets and SNOMED-CT identified
Data Preprocessing	Planned	ECG standardisation and preprocessing
Reasoning & CKP Framework	Planned	ECG-only and CKP paths defined
Grounding & Evaluation	Planned	Grounding, AURDC, and RFB defined
Testing	Planned	Module and integrated testing
Documentation	In Progress	Report and technical documentation

Testing and Validation

Test ID	Test Scenario	Expected Result	Status
TC-01	Progressive Lead Masking	Generates 12–1 lead inputs without reconstruction.	Planned
TC-02	ECG-only vs. CKP Reasoning	Compares both paths under the same masking levels.	Planned
TC-03	Grounding Verification	Verifies claims using available or derivable ECG evidence.	Planned
TC-04	CKP Relevance Control	Compares relevant and irrelevant CKP.	Planned
TC-05	Reasoning Failure Boundary	Identifies the unreliable reasoning lead level.	Planned
TC-06	Degradation Analysis	Computes the degradation curve, AURDC, and RFB.	Planned

Validation Method

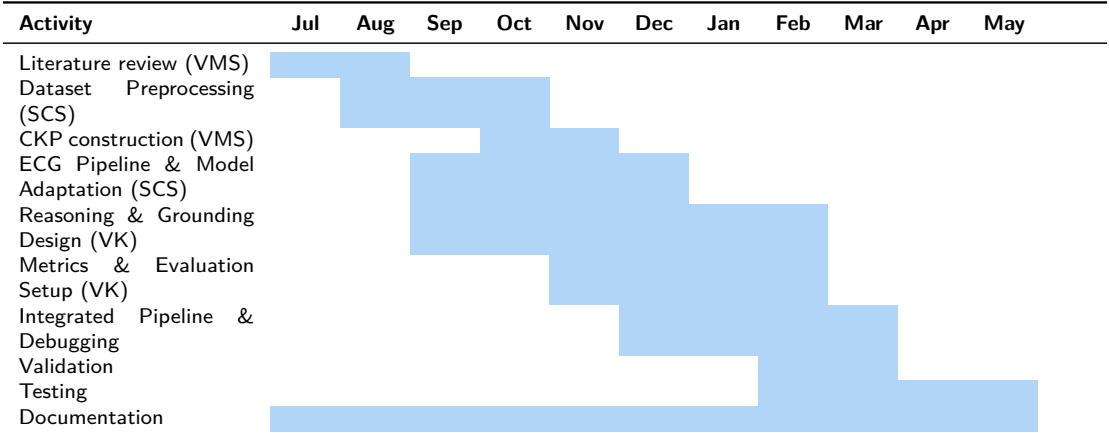
Component Validation: Each module produces outputs matching its defined input–output criteria.

Integration Testing: The complete ECG-to-grounding pipeline executes correctly without data-flow errors.

Statistical Validation: CKP-assisted performance is significantly compared with ECG-only across masking levels.

Cross-Dataset Validation: The framework maintains consistent evaluation trends across independent ECG datasets.

Project Timeline



Team Responsibilities

Team Member	Responsibilities	Current Status
Vedanth M S	Literature review, CKP design, methodology, presentation	In Progress
Siddharth S	Dataset preparation, preprocessing, lead masking, ECG model and representation	Planned
Vijay K	Reasoning pipeline, grounding, evaluation metrics, experiments and analysis	Planned
All Members	Validation, Testing, Documentation	Planned

Challenges, Risks, and Mitigation

Risk / Challenge	Possible Impact	Mitigation
Derived ECG lead handling complexity	Incorrect evidence availability assessment	Track direct and derivable ECG evidence using lead relationships
CKP construction complexity	Limited benefit or unsupported knowledge influence	Use ECG-relevant concepts and relationships from SNOMED-CT
Evidence Schema & hypothesis space complexity	Incorrect claim verification or candidate selection	Define standardized evidence rules and controlled hypothesis generation
Dataset variability	Inconsistent results across datasets	Use standardized preprocessing and external validation
Integration complexity	Schedule delay	Develop and test modules independently before integration
Computational resource	Longer experimental execution	Optimize inference and use GPU-based resources

Conclusion and Future Work

Conclusion

Problem: How does ECG reasoning degrade as physiological evidence is lost?
Approach: Compare ECG-only and CKP-assisted reasoning under progressive lead masking.
Current work: Framework, datasets, CKP, and grounding approach established.
Evaluation: Measure grounding quality and diagnostic performance across lead levels.
Outcome: Compute AURDC and identify the Reasoning Failure Boundary (RFB).

Future Work

- **Broaden disease coverage:** Evaluate more ECG conditions and categories.
- **Expand validation:** Test across more datasets and diverse patient populations.
- **Improve CKP coverage:** Add more ECG findings, relationships, and evidence rules.
- **Extend degradation analysis:** Study finer-grained and alternative lead-masking patterns.

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Thank You

Questions and Discussion

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